Sharpe-ratio variance inflation in entity-time financial panels

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Abstract

Entity-time financial panels are often evaluated by treating selected rows as independent observations, even when the economically meaningful object is a date-level portfolio return. This paper studies the resulting sampling-boundary problem for Sharpe-ratio inference. It defines a Sharpe Unified Variance Inflation Factor that compares long-run date-level delta-method variance with row-pooled IID variance, and pairs the diagnostic with an audit protocol using date aggregation, HAC-delta inference, dependent bootstrap resampling, researcher-menu correction, placebo tests, factor-alpha checks, and turnover-scaled cost sensitivity. Public Kenneth French and AQR benchmarks illustrate the protocol in momentum and factor-screen settings. In dependent null designs, row-naive rejection reaches about 35 percent at a nominal 5 percent level, while date-level HAC and bootstrap procedures remain closer to nominal size. The contribution is a reproducible financial econometrics testing framework for panel-based performance evidence.

Keywords: Sharpe ratio; financial econometrics; panel data; cross-sectional dependence; serial dependence; HAC inference; dependent bootstrap; multiple testing; portfolio signal evaluation; information redundancy
JEL Codes: C12; G11; G17

1 Introduction

1.1 Problem: The wrong sampling boundary

Empirical finance increasingly evaluates signals in entity-time panels: factor screens, anomaly sorts, automated portfolio-selection pipelines, model-selection grids, and trading-strategy backtests. A typical panel can be written as $\mathcal{P} = \{(t, i, s_{t,i}, r_{t+1,i})\}$, where t indexes dates, i indexes assets or portfolios, $s_{t,i}$ is a signal, and $r_{t+1,i}$ is the next-period return. One evaluation failure mode is to evaluate a positive edge using all selected symbol-date rows as though they were independent. This turns a panel with T dates and n_t selected rows per date into a nominal sample of $\sum_t n_t$, even when same-date rows share common shocks. This row-level calculation is used here as a baseline reference for bias quantification and as a concrete failure mode for automated signal-evaluation pipelines.

The issue extends beyond a single trading-strategy application. Factor screens, anomaly panels, model-selection procedures, and automated portfolio-signal evaluations can all multiply observations across assets on the same date while the economic payoff is realized through a date-level portfolio return. The paper uses public trading-panel and factor-screen examples because they make the sampling boundary visible, but the statistical problem is a general financial-econometrics question: how much Sharpe-ratio evidence remains after cross-sectional row multiplication and serial dependence in the date-return series are recognized?

The problem is not only a matter of notation. A row-pooled t-test implicitly treats the cross-section on a date as many independent draws from the same experiment. A portfolio signal or trading strategy, however, is marked to market by date. If an evaluation selects several assets or portfolios on the same date, the economically relevant object is the date-level portfolio return after aggregation. Cross-sectional dependence determines how much extra information the additional selected rows actually contain, and serial dependence determines the long-run variance of the date series. Standard clustering techniques, including the finance-panel guidance of Petersen (2009), address standard errors for linear regression coefficients or means. They do not by themselves define a complete inference record for nonlinear performance metrics such as the Sharpe ratio when the same data contain both cross-sectional shocks and serial memory. The missing object is a single variance-inflation accounting that starts from the row-pooled benchmark, moves through the date-return boundary, and ends at the long-run variance of the Sharpe functional.

Modern signal-search pipelines can evaluate thousands of entity-date rows and hundreds of menu variants in a single run. The computational ease of row-level evaluation makes the sampling-boundary error easier to automate than to detect. The threat model is therefore concrete: the problem is largest when multiple same-date entities are selected, returns share common shocks, the performance statistic is nonlinear, the researcher searches over menus, date returns have serial dependence, and trading or implementation costs are nontrivial.

Two terms are used throughout the paper. The Sharpe Unified Variance Inflation Factor, or Sharpe UVIF, is

$$\mathrm{UVIF}_{\mathrm{SR}} = \frac{\mathbb{V}\mathrm{ar}_{LR}(\widehat{\mathrm{SR}}_{date})}{\mathbb{V}\mathrm{ar}_{IID,row}(\widehat{\mathrm{SR}}_{row})}.$$

It is best interpreted as a sampling-boundary distortion ratio for reported Sharpe uncertainty. The numerator is the uncertainty of the economically relevant date-level Sharpe claim. The denominator is the uncertainty that would be reported by a row-pooled IID analysis of the selected entity-date returns. When the row-pooled and date-level estimators target the same Sharpe functional under exchangeability and equal-weight aggregation, Sharpe UVIF reduces to a design-effect variance inflation factor for the nonlinear Sharpe ratio. More generally, it is a target-aware audit bridge: date-level HAC inference tells the researcher what to believe, while Sharpe UVIF shows how much of the original row-pooled precision was manufactured by the wrong sampling boundary. The Effective Row Information Retention (ERIR) is a design-effect retention diagnostic; for example, ERIR = 0.11 means roughly 89 percent of nominal row-level information is redundant under the equicorrelation approximation.

1.2 Solution: Dependence-aware inference

This paper makes three contributions. First, it defines Sharpe UVIF, a variance-ratio diagnostic that links row-pooled IID Sharpe inference to date-level long-run Sharpe inference. Unlike mean-only design effects, Sharpe UVIF applies the delta method to the nonlinear Sharpe functional and therefore carries the long-run covariance of both returns and squared returns into the uncertainty calculation. Second, it embeds this variance accounting in a registry-defined evaluation protocol that separates cross-sectional row multiplication from serial dependence in date-level portfolio returns. Third, it documents the practical size distortion created by row-naive inference in public benchmark panels and Monte Carlo designs, showing that apparent precision can be produced by treating same-date dependent rows as independent evidence.

The empirical evidence is a public benchmark and stress-test campaign. It asks which familiar momentum, value, quality, beta, and size/book-to-market claims survive once the sampling boundary, dependence, bootstrap robustness, researcher menu, placebo, factor-alpha, and cost boundaries are imposed. The public Kenneth French and AQR data are rendered into the manuscript from production CSV artifacts. The conservative empirical question is whether apparently significant portfolio-signal claims are artifacts of row-level sampling and disappear after date aggregation, dependence adjustment, and researcher-menu correction. The same code also runs a Monte Carlo study comparing row-naive testing with date-level bootstrap, HAC-delta, block bootstrap, stationary bootstrap, and joint resampling decisions.

2 Related work

This paper sits at the intersection of literatures that are often discussed separately. Survey and grouped-error design effects explain why row counts overstate information (Kish, 1965; Moulton, 1986), and finance-panel standard error work explains why cross-sectional and time clustering matter in empirical finance (Petersen, 2009). Those results are central background, but they mainly address means or regression coefficients rather than non-linear portfolio-performance evidence after entity-date rows have been aggregated into a date-level return.

Sharpe-ratio inference follows a separate time-series literature. Lo (2002) studies Sharpe-ratio sampling under return dynamics, and robust Sharpe testing builds on HAC and bootstrap methods (Ledoit and Wolf, 2008; Opdyke, 2007; Andrews, 1991; Kiefer and Vogelsang, 2005; Kunsch, 1989; Politis and Romano, 1994; Lahiri, 2003). These tools define the date-level sampling law once the return series is specified. They do not quantify how much apparent Sharpe precision was created before the evidence was moved from selected entity-date rows to the economically relevant date-return boundary.

Researcher-menu and factor-zoo work addresses a third source of false precision. White Reality Check and Romano-Wolf procedures control data-snooping and familywise error in menus (White, 2000; Romano and Wolf, 2005), while the empirical asset-pricing literature documents false discoveries, factor-alpha fragility, anomaly decay, and trading costs (Harvey et al., 2016; Harvey and Liu, 2020a,b; Bailey and Lopez de Prado, 2014; Holm, 1979; Fama and French, 1993; Carhart, 1997; Fama and French, 2010; McLean and Pontiff, 2016; Almgren and Chriss, 2001; Hasbrouck, 2009; Frazzini et al., 2015). This paper is adjacent to that literature but distinct: its object is the sampling unit and return boundary before a row-level Sharpe claim is interpreted as portfolio-performance evidence.

3 Materials and methods

The protocol is a dependence-aware inference framework rather than a single test. Its first channel is cross-sectional: all evidence is translated from selected symbol-date rows into one economically tradeable date return, so same-date clusters cannot create sample size by row multiplication. Its second channel is serial: all Sharpe decisions are made on the date-return series using HAC-delta and dependent-bootstrap procedures. A candidate must survive both channels before the paper asks the separate researcher-menu, placebo, factor-alpha, and cost questions.

Operationally, the audit has three reported scales. Row-naive statistics are retained only as the baseline to be audited. Date-level Sharpe statistics are the objects used for confirmatory inference. Sharpe UVIF and ERIR translate the gap between those two scales into reported uncertainty and design-effect retention terms, respectively.

3.1 Signals and executability

For lookback L and skip q, the skipped compounded signal is

$$s_{t,i}^{(L,q)} = \prod_{j=t-L-q+1}^{t-q} (1 + r_{j,i}) - 1. \tag{1}$$

The bounds in (1) are ascending. With $q \geq 1$, the signal excludes the outcome return and is executable at date t when all signals in the available universe A_t are known before the trading decision. This simultaneity condition is automatic for the public Kenneth French portfolio panel because all signals are computed from lagged public returns. It is a required condition for applying the protocol to any other prediction panel.

The sign convention is $\operatorname{sign}(0) = 0$. Zero-signal and missing-signal rows receive zero economic weight and are excluded from threshold selection. The signed one-period row return is

$$y_{t+1,i}^{(L,q)} = \operatorname{sign}(s_{t,i}^{(L,q)}) r_{t+1,i}. \tag{2}$$

The confidence score is the within-date percentile rank of $|s_{t,i}^{(L,q)}|$. This percentile score is a baseline screening heuristic, not an optimal weighting theorem. It is scale-free, transparent, and invariant to monotone transformations of the signal magnitude. Z-scores, volatility-scaled scores, and learned weights are valid alternatives only if they are declared before the confirmatory run or included in the researcher menu. The protocol's main object is the sampling boundary, not weighting optimality. With N_t finite non-zero signals and average midrank $R_{t,i}$,

$$c_{t,i}^{(L,q)} = \begin{cases} \dfrac{R_{t,i} - 1}{N_t - 1}, & N_t > 1, \\ 1, & N_t = 1. \end{cases} \tag{3}$$

For dates with $N_t = 1$, the convention $c_{t,i} = 1$ represents a single-asset conviction rather than an undefined percentile rank. Dates with $N_t = 0$ are inactive for that menu element. Ties use average midranks. When $N_t = 2$, the available confidence scores are $(0, 1)$, or $(0.5, 0.5)$ under a tie, so high threshold decisions are coarse. The empirical tables therefore report selected counts by threshold rather than treating the threshold scale as equally granular across dates.

3.2 Date returns and effective sample size

For threshold τ, selected set $\mathcal{S}_t(L,q,\tau) = \{i : c_{t,i}^{(L,q)} \geq \tau\}$, and optional non-negative weights $a_{t,i}$, the date return is

$$\bar{r}_{t+1}(L,q,\tau) = \frac{\sum_{i\in\mathcal{S}_t} a_{t,i} y_{t+1,i}^{(L,q)}}{\sum_{i\in\mathcal{S}_t} a_{t,i}}. \tag{4}$$

The denominator is the total active economic weight on date t, not a generic row count. Under the primary equal-weighted evaluation $a_{t,i} = 1$, it reduces to the number of selected rows. Confidence-weighted and signal-magnitude-weighted returns are sensitivity checks.

For descriptive diagnostics, the artifacts report a positive-sequence effective sample size for the finite date-return series. The autocorrelation sum is truncated at the first negative sample autocorrelation, up to a fixed maximum lag. This is not a substitute for HAC or bootstrap inference; it is a compact diagnostic for the loss of information from serial dependence and is not used as a decision rule.

3.3 Sharpe inference

The protocol separates estimation from testing. It reports two-sided 95 percent intervals for $\widehat{\mathrm{SR}}$ and one-sided positive-edge p-values for $H_0 : \mathrm{SR} \leq 0$. Bootstrap inference resamples the date-return series by IID date, moving block, or stationary bootstrap. The positive-edge bootstrap p-value uses one-draw smoothing:

$$p^+ = \frac{1 + \#\{\widehat{\mathrm{SR}}_b^* \leq 0\}}{B+1}. \tag{5}$$

For HAC inference, reindex the finite realized date-return observations as R_u, $u = 1,\ldots,T_R$, where R_u is the strategy excess return for the decision date after subtracting the applicable risk-free rate when the source is not already a long-short excess-return series. Let $g_u = (R_u, R_u^2)'$, $\hat{\mu} = T_R^{-1}\sum_u R_u$, $\hat{m}_2 = T_R^{-1}\sum_u R_u^2$, and $\hat{v} = \hat{m}_2 - \hat{\mu}^2$. The denominator uses the T_R-moment variance, not the $T-1$ bias correction, because it is the plug-in moment entering the smooth functional

$$f(\mu, m_2) = \sqrt{252}\frac{\mu}{(m_2 - \mu^2)^{1/2}}.$$

Following Ledoit and Wolf (2008), the Sharpe ratio is treated as a smooth functional $f(\theta)$ of the moment vector $\theta = \mathbb{E}[g_t]$. On sets where $m_2 - \mu^2$ is bounded away from zero, this map is Hadamard differentiable. The asymptotic normality of $\sqrt{T_R}(\widehat{\mathrm{SR}} - \mathrm{SR})$

follows from the functional delta method provided the long-run covariance matrix Ω is consistently estimated by a kernel-based HAC estimator. Evaluated at $(\hat{\mu}, \hat{m}_2)$,

$$\widehat{\mathrm{SR}} = \sqrt{252}\frac{\hat{\mu}}{\sqrt{\hat{v}}}, \qquad \nabla f(\hat{\mu}, \hat{m}_2) = \left(\sqrt{252}\frac{\hat{m}_2}{\hat{v}^{3/2}}, -\frac{1}{2}\sqrt{252}\frac{\hat{\mu}}{\hat{v}^{3/2}}\right)'.$$

With $\widehat{\Omega}$ the Bartlett-kernel HAC estimate of the long-run covariance of g_u, the HAC-delta variance is

$$\widehat{\mathbb{V}\mathrm{ar}}(\widehat{\mathrm{SR}}) = \nabla f(\hat{\mu}, \hat{m}_2)'\widehat{\Omega}\nabla f(\hat{\mu}, \hat{m}_2)/T_R. \tag{6}$$

Explicitly, for centered moments $\tilde{g}_u = g_u - \bar{g}$,

$$\widehat{\Gamma}_k = T_R^{-1}\sum_{u=k+1}^{T_R} \tilde{g}_u\tilde{g}'_{u-k}, \qquad \widehat{\Omega} = \widehat{\Gamma}_0 + \sum_{k=1}^{K_T}\left(1 - \frac{k}{K_T+1}\right)\left(\widehat{\Gamma}_k + \widehat{\Gamma}'_k\right).$$

This is a covariance matrix for the joint process (R_u, R_u^2), so the off-diagonal terms carry the long-run covariance between the sample mean and second moment into the Sharpe-ratio delta method. The bandwidth K_T is a tuning choice and may depend on T. The implementation reports the selected bandwidth and uses a documented automatic Bartlett bandwidth as the default; important empirical claims should be checked against reasonable nearby bandwidths and against the block-bootstrap intervals rather than relying on a single HAC lag choice. The technical appendix reports prewhitened HAC and fixed-fraction HAC sensitivity checks.

Because financial date returns can be heavy-tailed, the HAC-delta calculation is not allowed to stand alone. The manuscript reports bootstrap p-values and intervals adjacent to HAC estimates; when the bootstrap evidence is materially weaker than the HAC evidence, the candidate is treated as robustness-fragile rather than as a clean positive-edge finding.

3.4 Researcher menus, placebos, and factor alpha

For a fixed menu $\mathcal{M}$ of lookbacks and thresholds, the primary joint adjustment is Romano-Wolf stepdown max-t resampling over $\{R_u(m) : m \in \mathcal{M}\}$. Holm-adjusted marginal bootstrap p-values are secondary. The protocol also reports White's Reality Check for the maximum mean return and the Deflated Sharpe Ratio diagnostic of Bailey et al. (2014). In the public campaign, the menu is the finite registry-defined set

$$\mathcal{M} = \{(c, \theta_c, \tau) : c \in \mathcal{C},\ \tau \in \mathcal{T}_c\},$$

where θ_c denotes the candidate's frozen signal definition. For panel candidates, $\mathcal{T}_c = \{0.0, 0.3, 0.5, 0.7, 0.9\}$ and any lookback and skip values are fixed in `code/campaign_` `registry.json` rather than selected after seeing results. Pre-aggregated AQR factor-series candidates use the singleton threshold menu $\mathcal{T}_c = \{0.0\}$; because they do not contain row-level constituent signals, same-date panel permutation is not applicable to them. The reason is mechanical. If N independent strategies with zero true Sharpe are searched over a sample of length T, the maximum observed daily Sharpe is on the order of $\sqrt{2\log(N)/T}$ before any economic signal exists. Real researcher menus are usually correlated rather than independent, so the protocol uses Romano-Wolf stepdown to control familywise error while preserving the joint dependence among menu elements.

The fixed-menu condition is a confirmatory convention, not a claim that research is never iterative. In practice, exploratory work should be separated from the final evaluation by freezing the signal definitions, lookbacks, thresholds, filters, cost assumptions, and benchmark model before the confirmatory run. If discarded variants influenced the final choice, they belong in the researcher menu or the exercise should be reported as exploratory. A stronger design uses a locked holdout period or an independently refreshed sample after the menu is frozen.

The public placebo is a same-date permutation design. For each date, the vector of signals is randomly permuted across portfolios, preserving the date's cross-sectional signal distribution and missingness while breaking the signal-return pairing. The signed next-day returns, confidence ranks, selected counts, turnover, and date aggregation are then recomputed. Repeating this procedure gives a permutation null for the observed public momentum panel.

This placebo is a test under a within-date exchangeability hypothesis: under the null, the assignment of signals $s_{t,i}$ to next-period returns $r_{t+1,i}$ is exchangeable within each date t. If signals are structurally tied to size, sector, liquidity, country, exchange, beta, or other common-factor state variables, an unrestricted same-date permutation can break the wrong null. In those cases the protocol should permute within registry-defined blocks, residualize returns against the structural variables before permutation, or mark the placebo as not applicable. The permutation gate is therefore a design-specific diagnostic rather than a universal validity guarantee.

The factor-alpha test estimates

$$R_t = \alpha + \beta_M(MKT_t - RF_t) + \beta_S SMB_t + \beta_H HML_t + \beta_U MOM_t + \varepsilon_t, \tag{7}$$

using HAC standard errors. The null is $H_0 : \alpha \leq 0$. Including momentum is essential because the public signal is itself momentum-like.

3.5 Turnover-scaled costs

For the equal-weighted threshold portfolio, define target weights

$$w_{t,i} = \frac{\operatorname{sign}(s_{t,i})\mathbf{1}\{i \in \mathcal{S}_t\}}{|\mathcal{S}_t|}, \qquad w_{t,i} = 0 \text{ if } |\mathcal{S}_t| = 0. \tag{8}$$

Daily turnover is the half-L^1 distance between adjacent target weight vectors,

$$TO_t = \frac{1}{2}\sum_i |w_{t,i} - w_{t-1,i}|, \qquad \widehat{TO} = T_{TO}^{-1}\sum_t TO_t, \tag{9}$$

where the average is over adjacent dates with at least one active side. For cost c per full rebalance in decimal units, the cost-adjusted Sharpe ratio is

$$\widehat{\mathrm{SR}}_{net}(c) = \frac{\hat{\mu} - \widehat{TO}\,c}{\hat{\sigma}}\sqrt{252}. \tag{10}$$

This is a first-order cost stress. It assumes linear cost per unit of turnover and holds the return volatility fixed after subtracting costs. Those assumptions are reasonable for a screening evaluation of diversified, low-frequency panels, and the resulting cost drag should be read as a lower bound when liquidity demand is material. They are not a full execution model for high-frequency, capacity-constrained, or illiquid strategies. Such applications require spread, impact, latency, borrow, and capacity models before any tradability claim. The protocol establishes a statistical lower bound for viability; it does not account for liquidity-driven alpha decay at larger assets under management. The empirical artifacts also report the break-even cost in basis points, defined by $\hat{\mu} - \widehat{TO}c = 0$. A square-root impact capacity extension in the spirit of Almgren and Chriss (2001) and the broader market-impact literature (Bouchaud et al., 2008) is provided in the technical appendix. The public sorted-portfolio examples do not contain average daily volume or order-size information, so they do not support a capacity claim.

3.6 Composite viability decision

The protocol reports component diagnostics, but the final claim boundary is deliberately stricter than any single statistic. The gates are a lexicographic filter: a candidate must first clear the date-level statistical boundary, then the dependent-bootstrap robustness boundary, then the researcher-menu and placebo integrity boundaries, and finally the economic cost boundary. A candidate is called viable only if it clears a registry-defined composite gate:

$$p_{\text{boot}} \le \alpha, \qquad p_{\text{perm}} \le \alpha, \qquad p_{\text{HAC}} \le \alpha, \qquad p_{\text{RW}} \le \alpha, \qquad \widehat{\mathrm{SR}}_{net}(c) > 0.$$

This is a conservative policy rule, not an optimal weighted test or a claim that the gates exhaust all possible failure modes. The gates are conjunctive because each one addresses a different reason a trading claim can fail: bootstrap-fragile Sharpe evidence, a placebo-consistent signal, dependence-adjusted insignificance, researcher-menu selection, or nonpositive net economics. Applications with different decision costs may pre-specify different gates, but they should not choose the gate sequence after seeing the results. The reported artifacts make the gate threshold reproducible: in addition to the conventional $\alpha = 0.05$, a reader can recompute pass counts at stricter levels such as $\alpha = 0.01$ without changing the statistical objects being tested. The same-date permutation gate is a panel placebo: it is applicable only when row-level signals can be permuted within date. Pre-aggregated factor series can still be economically interesting and can pass HAC-delta or Romano-Wolf tests, but they do not contain the row-level signal panel needed for this placebo. Such cases are reported as not applicable for the permutation gate rather than as passed panel evaluations.

The stages have different interpretations. The permutation gate asks whether the signal-return pairing beats a same-date placebo. The HAC gate asks whether the resulting date-return Sharpe is positive after serial dependence and heteroskedasticity are accounted for. The bootstrap gate asks whether that conclusion survives resampling of the date-return series. The Romano-Wolf gate asks whether the claim survives the fixed researcher menu. The cost gate asks whether the edge remains economically usable after turnover-scaled frictions. Factor-alpha, holdout, subperiod, and stationarity diagnostics are reported alongside these gates to prevent a single favorable statistic from becoming the claim.

3.7 Sampling-boundary theory

Assumption 1 (Equicorrelated row model)**.** *For each date t, selected row returns satisfy*

$$y_{t,i} = \mu + \lambda_t + \epsilon_{t,i},$$

where λ_t is a common date shock, $\epsilon_{t,i}$ is idiosyncratic, λ_t is uncorrelated with $\epsilon_{t,i}$, $\mathbb{V}\mathrm{ar}(\lambda_t) = \sigma_\lambda^2$, and $\mathbb{V}\mathrm{ar}(\epsilon_{t,i}) = \sigma_\epsilon^2$. The selected count is $n_t = \bar{n}$ for the displayed derivation, and dates are independent.

Under this model,

$$\sigma^2 = \mathbb{V}\mathrm{ar}(y_{t,i}) = \sigma_\lambda^2 + \sigma_\epsilon^2, \qquad \rho = \mathbb{C}\mathrm{orr}(y_{t,i}, y_{t,j}) = \frac{\sigma_\lambda^2}{\sigma_\lambda^2 + \sigma_\epsilon^2} \quad (i \neq j).$$

Thus ρ is the intraclass correlation induced by the common date shock; economically, it is the fraction of row-return variance explained by shocks shared by same-date selections.

In empirical diagnostics, ρ is estimated after the candidate threshold is fixed. For each panel candidate, the selected signed row returns are aligned by date and asset or portfolio, and $\hat{\rho}$ is the average off-diagonal sample correlation among active row-return series with overlapping dates; $\bar{n}$ is the average selected count across active dates. When the selected set varies over time, this equicorrelation estimate is a descriptive compression of the same-date covariance structure, not an identifying assumption for the confirmatory HAC or bootstrap tests.

Proposition 1 (Mean-return design effect). *Under the equicorrelated row model, the ratio of the naive row-pooled reported standard error of the mean to the date-level standard error is*

$$\frac{se_{row,naive}}{se_{date,true}} = \{1 + (\bar{n} - 1)\rho\}^{-1/2}.$$

For $\rho > 0$ and $\bar{n} > 1$, the naive row-level report is too small.

This calculation is not new as a sampling result; it is the Moulton-like special case of the paper's UVIF logic, written for signed trading-panel rows and date-level portfolio returns. To avoid conflict with the researcher menu notation $\mathcal{M}$, write the cross-sectional variance design effect as

$$\mathrm{UVIF}_{\mu,xs}(\bar{n}, \rho) = 1 + (\bar{n} - 1)\rho.$$

The square root is reported when the object is a standard-error or t-statistic adjustment:

$$\mathrm{UVIF}^{1/2}_{\mu,xs}(\bar{n}, \rho) = \sqrt{1 + (\bar{n} - 1)\rho}.$$

Thus UVIF denotes a variance ratio, while $\mathrm{UVIF}^{1/2}$ denotes the associated standard-error multiplier. This mean-return calculation is expository: it makes the scale of understatement visible before the paper turns to HAC and bootstrap decisions on the realised date-return series. The magnitude can be economically large. The relevant inflation factor for the true standard error relative to the naive row report is $\mathrm{UVIF}^{1/2}_{\mu,xs}(\bar{n}, \rho)$. With $\bar{n} = 50$, increasing same-date correlation from $\rho = 0.10$ to $\rho = 0.60$ raises this factor from 2.43 to 5.51. Thus the largest damage occurs precisely in regimes where many names are selected and common shocks dominate idiosyncratic noise, which is the pattern one expects during market stress.
The proof is given in the technical appendix.

Remark 1 (Heterogeneous factor covariance). *The equicorrelated model is a closed-form illustration, not a restriction imposed by the protocol. Let a same-date signed-return vector satisfy*

$$y_t = \mu\mathbf{1} + Bf_t + u_t,$$

where B contains asset-specific loadings, $\mathbb{V}\mathrm{ar}(f_t) = \Sigma_f$, and $\mathbb{V}\mathrm{ar}(u_t) = D_t$. Then the *same-date covariance matrix is*

$$\Sigma_t = B\Sigma_f B' + D_t,$$

with heterogeneous pairwise correlations whenever loadings or idiosyncratic variances differ. For any displayed portfolio weights w_t, the relevant one-date variance is $w_t'\Sigma_t w_t$, not the row-independent variance. The protocol avoids estimating the full Σ_t as a prerequisite for inference by moving the evidence to the realised date-return series and then using HAC or dependent bootstrap inference on that series. The design-effect calculation is therefore best read as the transparent special case that explains why row pooling fails.

3.8 Unified variance inflation for the Sharpe ratio

The mean-return design effect is useful but incomplete because the reported performance metric is usually the Sharpe ratio. The Sharpe ratio is a nonlinear functional of the first two moments, so its variance inflation depends on the covariance of R_t and R_t^2, not only on the variance of R_t. This motivates the Unified Variance Inflation Factor:

$$\mathrm{UVIF}_{\mathrm{SR}} = \frac{\mathbb{V}\mathrm{ar}_{LR}(\widehat{\mathrm{SR}}_{portfolio})}{\mathbb{V}\mathrm{ar}_{naive}(\widehat{\mathrm{SR}}_{row})}. \tag{11}$$

The numerator is the long-run delta-method variance at the date-return boundary. The denominator is the row-pooled delta-method variance that would be reported if the selected rows were treated as independent observations. Equation (11) is therefore a reported-uncertainty distortion ratio. It compares the uncertainty of the economically relevant date-level Sharpe claim with the uncertainty that would be reported under row-pooled IID analysis. Under exchangeability of the linearized Sharpe influence function, and when the row-pooled and date-level estimators target the same equal-weight Sharpe functional, the cross-sectional part of this variance ratio reduces to the same $1 + (\bar{n} - 1)\rho$ design effect shown above, with ρ applied to the linearized Sharpe influence rather than to raw returns. Outside that special case, Sharpe UVIF is target-aware: it measures the gap between the row-level report and the date-level claim that can be traded, allocated to, or interpreted as portfolio evidence.

Proposition 2 (Sharpe UVIF decomposition)**.** *Let R_t be the date-level portfolio excess return and $G_t = (R_t, R_t^2)'$. Define $\theta_R = (\mu_R, m_{2,R})' = \mathbb{E}[G_t]$, $v_R = m_{2,R} - \mu_R^2 > 0$, and*

$$d_R = \nabla f(\theta_R) = \left(\sqrt{252}\frac{m_{2,R}}{v_R^{3/2}}, -\frac{1}{2}\sqrt{252}\frac{\mu_R}{v_R^{3/2}}\right)'.$$

Let $\Gamma_k^R = \mathbb{C}\mathrm{ov}(G_t, G_{t-k})$ and $\Omega_R = \Gamma_0^R + \sum_{k\geq 1}(\Gamma_k^R + \Gamma_k^{R\prime})$. For the selected row return $Y_{t,i}$, let $H_{t,i} = (Y_{t,i}, Y_{t,i}^2)'$, $\theta_Y = \mathbb{E}[H_{t,i}]$, $d_Y = \nabla f(\theta_Y)$, and $\Psi_Y = \mathbb{V}\mathrm{ar}(H_{t,i})$. With a constant *selected count $\bar{n}$, the first-order Sharpe variance inflation relative to a row-pooled IID report is*

$$\mathrm{UVIF}_{\mathrm{SR}} = \underbrace{\frac{\bar{n}\, d_R' \Gamma_0^R d_R}{d_Y' \Psi_Y d_Y}}_{\textit{cross-sectional UVIF component}} \times \underbrace{\frac{d_R' \Omega_R d_R}{d_R' \Gamma_0^R d_R}}_{\textit{Sharpe long-run serial factor}} . \tag{12}$$

Moreover, the serial factor can be written as

$$1 + 2\sum_{k=1}^{\infty} \psi_k^{\mathrm{SR}}, \qquad \psi_k^{\mathrm{SR}} = \frac{d_R' \Gamma_k^R d_R}{d_R' \Gamma_0^R d_R},$$

and the scalar long-run variance entering the numerator is

$$d_R' \Omega_R d_R = \frac{252}{v_R^3} \left[m_{2,R}^2 \Omega_{11} - \frac{\mu_R m_{2,R}}{2} (\Omega_{12} + \Omega_{21}) + \frac{\mu_R^2}{4} \Omega_{22} \right], \tag{13}$$

where Ω_{ab} denotes the (a, b) element of Ω_R.

The proof is given in the technical appendix.

Remark 2 (Connection to Moulton-like design effects)**.** *If the linearized row Sharpe influence $\zeta_{t,i}^Y = d_Y'(H_{t,i} - \theta_Y)$ is exchangeable within date with intraclass correlation ρ_ζ, and the date-level linearization is the equal-weighted average of those row influences, the cross-sectional factor in* (12) *reduces to* $1 + (\bar{n} - 1)\rho_\zeta$*. The mean-return design effect is the special case in which the influence variable is $Y_{t,i} - \mu$. The Sharpe case replaces raw return correlation by correlation of the linearized nonlinear-performance influence, which is why skewness, kurtosis, and volatility clustering enter the variance inflation.*

3.9 Higher-moment UVIF and non-normality

The two-moment Sharpe UVIF already depends on fourth moments through the variance of R_t^2. If a venue or application uses an adjusted Sharpe ratio that explicitly corrects for skewness and kurtosis, the same delta-method construction extends the moment vector from $(R_t, R_t^2)'$ to $(R_t, R_t^2, R_t^3, R_t^4)'$. The resulting higher-moment UVIF is the ratio of the adjusted-Sharpe long-run variance at the date-return boundary to the corresponding row-pooled IID variance. It is reported as a tail-risk diagnostic, not as an additional rejection gate. The full expression is in the technical appendix.

3.10 Effective row information retention

For interpretability, the paper reports Effective Row Information Retention (ERIR) as a design-effect diagnostic rather than as an additional hypothesis test or formal information claim. Under the equicorrelated row model, the cross-sectional mean-return design effect is

$$DE_{\mu,xs} = 1 + (\bar{n} - 1)\rho. \tag{14}$$

The corresponding row-retention diagnostic is therefore

$$\text{ERIR} = \{1 + (\bar{n} - 1)\rho\}^{-1}. \tag{15}$$

ERIR is interpreted as the fraction of nominal row-level information that remains nonredundant under the equicorrelation approximation. For example, ERIR $= 0.111$ means that approximately $1 - 0.111 = 0.889$, or 89 percent, of the nominal row-level information implied by naive row pooling is redundant under this diagnostic approximation. The statistic is descriptive only. Confirmatory inference is conducted using date-level HAC-delta and dependent-bootstrap procedures.

Assumption 2 (Fixed-menu bootstrap validity)**.** *For each fixed menu element m, the date-return series $\{\bar{r}_t(m)\}$ is strictly stationary and alpha-mixing with moments of order $4+\delta$. The long-run covariance of $(\bar{r}_t, \bar{r}_t^2)$ is finite and non-degenerate, and $\mathbb{V}\text{ar}(\bar{r}_t) \geq \eta > 0$. The block length satisfies $\ell \to \infty$ and $\ell/T \to 0$. The researcher menu $\mathcal{M}$ is fixed and finite.*

Corollary 1 (Fixed-menu validity statement)**.** *Under the fixed-menu bootstrap-validity assumption, the HAC-delta and block-bootstrap procedures consistently estimate the marginal sampling law of $\widehat{\text{SR}}(m)$ for each fixed m. If marginal positive-edge p-values are valid and the stepdown rejection regions are monotone, Holm and Romano-Wolf stepdown control familywise error asymptotically for the fixed menu.*

Justification. This is an assembly of standard results rather than a new theorem. The Sharpe functional is Hadamard differentiable on the set $m_2 - \mu^2 \geq \eta$. A central limit theorem for the sample mean of $(\bar{r}_t, \bar{r}_t^2)$, the delta method, and standard block-bootstrap consistency for mixing sequences imply marginal Sharpe validity (Kunsch, 1989; Politis and Romano, 1994; Lahiri, 2003). For a fixed finite menu, Holm controls FWER for valid marginal p-values, and Romano-Wolf stepdown uses joint max-statistic resampling under the standard monotonicity requirement (Holm, 1979; Romano and Wolf, 2005).

Remark 3 (Scope of the validity statement)**.** *The validity statement does not cover long memory, infinite-variance heavy tails, growing researcher menus, structural breaks, or nonstationary date returns. The empirical evaluation reports stationarity diagnostics and stress simulations, but those outputs are caveats and diagnostics, not proof of asymptotic validity in those settings.*

Table 1: Synthetic positive-control audit.

Panel	$\bar{\rho}$	Adj.	SR	HAC p^+	RW p	Perm. p^+
Synthetic control	0.202	4.58	1.744	7.67e-13	5.00e-04	9.99e-04

4 Synthetic positive-control validation

The protocol is not mechanically anti-discovery. To check that the composite gates can pass a true signal, a synthetic panel is generated with a persistent predictive signal, moderate same-date dependence, and serial dependence in the date-level returns. The signal is constructed before inference is run, the same date-aggregation and audit gates are applied, and the resulting positive-control candidate passes HAC, Romano-Wolf, and same-date permutation checks. This validation is not used as evidence for a public trading strategy; it shows that the procedure can accept known positive evidence when the data-generating process contains a genuine date-level edge.

Appendix material. The worked numerical example, full proof details, positive-control data generation notes, and secondary robustness artifacts are reported in the technical appendix. The main manuscript keeps only the claim boundary and the evidence needed to evaluate it.

5 Public benchmark and stress-test evidence

The public experiment is a registry-defined candidate campaign rather than a single example. It evaluates the canonical Kenneth French winner-minus-loser momentum decile benchmark, a broader Kenneth French decile-rank threshold profile, Kenneth French 25 Size/BM portfolios with a skipped-momentum signal, and public AQR value, momentum, quality, beta, and HML-style factor candidates. The AQR series are economically meaningful factor-series benchmarks, but they are pre-aggregated series rather than row-level constituent panels; when the same-date permutation gate cannot be applied, the benchmark is outside the full panel-audit gate. The main tables therefore separate panel candidates from single-series factor benchmarks. Failed public-data access attempts are kept in the provenance artifacts rather than treated as main empirical results.

The generated empirical tables in this section are produced from a full-campaign root by `code/render_manuscript_artifacts.py`. Table 2 reports panel candidates subject to the full audit gate. Table 3 reports pre-aggregated factor benchmarks subject to time-series inference only. Table 4 verifies that the canonical WML candidate matches the direct French winner-minus-loser decile benchmark. Table 5 gives the dependence audit summary using row p-values, UVIF, ERIR, audit p-values, and net Sharpe

Table 2: Panel candidates subject to the full audit gate.

Panel candidate	SR	HAC p+	Boot p+	RW p	Perm p	Net SR	Pass 5%
French momentum deciles, rank-threshold panel	0.447	7.54e-05	2.00e-04	2.00e-04	1.000	0.447	no
French WML decile benchmark	0.497	5.53e-06	2.00e-04	2.00e-04	1.00e-04	0.497	yes
French 25 Size/BM, skipped-momentum panel	0.036	0.445	0.458	0.516	0.947	-0.072	no

Table 3: Single-series factor benchmarks subject to time-series inference only.

Single-series benchmark	SR	HAC p+	Boot p+	RW p	Scope
AQR Betting Against Beta, equity factors	0.741	3.08e-13	2.00e-04	2.00e-04	time-series only
AQR Quality Minus Junk	0.536	2.26e-05	2.00e-04	2.00e-04	time-series only
AQR HML Devil	0.336	0.004	0.002	0.002	time-series only
AQR Value and Momentum Everywhere	2.623	1.62e-04	2.00e-04	2.00e-04	time-series only

status for panel candidates. Table 6 reports the targeted $L = 1, 5, 21$ horizon-effect audit for the dynamic Size/BM momentum panel. The technical appendix reports the full secondary table set: sampling-boundary diagnostics, same-date permutation details, HAC bandwidth/prewhitening/fixed-b robustness, gate sensitivity, holdout splits, and cost sweeps.

The factor-alpha, data-snooping, stationarity, holdout, and cost tables are secondary diagnostics, not replacements for the primary date-return decision. A raw positive Sharpe that does not survive FF3 plus momentum alpha, bootstrap robustness, Romano-Wolf/White checks, same-date placebos, or plausible costs should not be interpreted as reliable portfolio-performance evidence. This is the empirical implication of the protocol: familiar public candidates can look attractive at one boundary and fail at the full decision boundary.

The panel-candidate table reports only sources with row-level signal-return structure and therefore eligibility for the full panel audit gate. The single-series benchmark table reports pre-aggregated factor returns separately; those rows are time-series benchmarks, not failed panel-audit candidates.

The benchmark validation table shows that the panel implementation matches the direct French winner-minus-loser construction.

The dependence audit table summarizes the boundary directly: row-level evidence can disappear once same-date redundancy, ERIR, placebo evidence, and turnover-scaled net Sharpe are evaluated together.

The horizon table shows that changing the lookback reduces turnover in the Size/BM stress panel but does not convert it into a robust net-Sharpe claim.

The campaign separates benchmark validation from broader search claims. The canonical French WML candidate is expected to match the direct Kenneth French winner-minus-loser decile construction and is reported as a public benchmark check. This resolves the apparent $\widehat{\mathrm{SR}} = 0.082$ benchmark problem in earlier artifacts: that number came from an earlier rank-threshold stress candidate, not from the canonical WML construction. In

Table 4: Canonical French momentum benchmark validation.

Candidate	thr	n	start	end	panel SR	direct WML SR	scale
French WML decile benchmark	0.900	26131	1926-11-04	2026-04-30	0.497	0.497	0.500

Table 5: Dependence audit summary for panel candidates.

Candidate	Row p+	UVIF	ERIR	Audit p+	Net SR	Status
French momentum deciles, rank-threshold panel	1.70e-04	1.000	1.000	1.000	0.447	Fails permutation gate
French WML decile benchmark	4.85e-04	1.000	1.000	2.00e-04	0.497	Passes composite gate
French 25 Size/BM, skipped-momentum panel	0.253	9.002	0.111	0.947	-0.072	Fails cost gate

Note: ERIR is a design-effect retention diagnostic and is not used as a rejection rule.

the rendered campaign, the WML panel Sharpe matches the direct WML Sharpe at approximately 0.50, and the scale difference is exactly one-half because the panel portfolio holds equal long and short legs while the direct WML series is winner minus loser. The broader rank-threshold profile and the French Size/BM dynamic-momentum panel are evaluated as stress cases rather than as canonical anomaly replications. The AQR factor-series candidates have positive time-series evidence in the reported HAC and bootstrap diagnostics, but because they are pre-aggregated factor returns rather than row-level signal panels, they cannot be promoted to passed panel-evaluation candidates.

The horizon-effect audit shows that the dynamic Size/BM momentum stress case fails because of weak signal and execution drag, not because the canonical momentum benchmark is missing. At threshold 0.5, $L = 1$ has very high turnover and a negative 5 bps net Sharpe; $L = 21$ lowers turnover substantially but still leaves net Sharpe slightly negative. The empirical Sharpe UVIF relative to row-naive inference is about 11–13 across the three horizons, while the serial-only UVIF is close to one in this sample. Thus the dominant inflation channel in this public stress panel is cross-sectional row multiplication; a longer lookback does not mechanically make serial dependence the dominant force. For the dynamic Size/BM panel in Table 5, the reported ERIR is 0.111. Under the equicorrelation diagnostic, this means $1 - 0.111 = 0.889$, so approximately 89 percent of the row-level information implied by naive row pooling is redundant. The dependence audit summary is intentionally artifact-backed rather than illustrative. It does not assert that every named factor is a panel candidate. Pre-aggregated AQR factor series are excluded from that table because they do not provide row-level signal-return pairs from which row p-values, same-date permutation tests, or panel ERIR can be computed.

6 Monte Carlo evidence

The Monte Carlo study is designed as a size-and-power audit of the sampling boundary, not as a search for a favorable strategy. Each replication generates a symbol-date return panel, applies the same date-aggregation rule used in the empirical protocol, and then

Table 6: Horizon-effect audit for the dynamic Size/BM momentum panel.

L	thr	dates	avg names	gross SR	net SR 5bps	turnover	UVIF SR	serial UVIF	HAC p+
1	0.500	3497	12.913	0.073	-0.394	0.733	11.498	0.891	0.386
5	0.500	3493	13.000	-0.312	-0.528	0.339	12.880	0.990	0.878
21	0.500	3477	13.000	0.058	-0.050	0.169	12.121	0.933	0.412

asks whether each inference method rejects a positive-edge null at the nominal 5 percent level. The base panel is

$$r_{t,i} = \mu_t + \beta_i' f_t + \epsilon_{t,i}, \qquad f_t = \Phi f_{t-1} + u_t.$$

The grid varies serial dependence, cross-sectional dependence, GARCH volatility, multifactor exposure, and regime switching. The high-persistence AR(1) case is reported as such; it is not labeled as a long-memory design. The high-volatility GARCH case is not described as an infinite-variance heavy-tail design. The production campaign uses 1000 Monte Carlo replications for the displayed size and rejection-rate tables, $T = 1000$ dates and $N = 50$ entities in the main null designs, 5000 bootstrap draws per simulation, and seed-indexed replications generated from the simulation index. Moving-block and stationary bootstrap procedures use the same block-length rule as the empirical audit code, and the full DGP grid, seed policy, registry hash, and run metadata are included in the reproducibility package. The null designs set the true edge to zero while varying cross-sectional correlation and serial memory; the power designs then increase the true annualized Sharpe on the same dependence-aware decision boundary. The important object is the gap between the row-naive rejection rate and the rejection rates of methods that operate on date-level returns. In the displayed null grid, the row-naive test overrejects around 35–37 percent under dependent nulls, while HAC-delta, moving-block bootstrap, stationary bootstrap, and Romano-Wolf remain much closer to nominal size. In the power grid, row-naive rejection reaches about 50 percent at a true annualized Sharpe near 0.19, before the dependence-aware methods show comparable power; that gap is false precision, not usable evidence.

The size and power experiment compares row-level naive testing, date-IID bootstrap, HAC-delta inference, moving-block bootstrap, stationary bootstrap, and Romano-Wolf joint resampling. The coverage experiment focuses on the date-level Sharpe target, so row-naive intervals are interpreted as coverage of the economically relevant date-level claim rather than coverage of a separate row-distribution Sharpe. The purpose is not to rank methods by power alone. A method that appears powerful because it overrejects under the null is not acceptable for confirmatory evaluation.

Table 7 gives the main size evidence. The row-naive column is the familiar symbol-date analysis; the HAC-delta, moving-block, stationary, and Romano-Wolf columns make

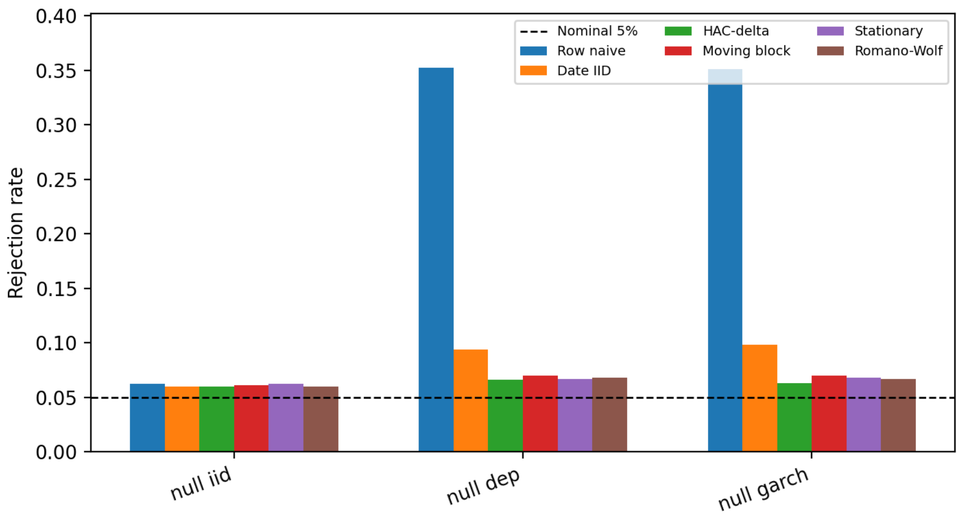

Figure 1: Null rejection rates under dependent and IID designs. The dashed line is the nominal 5 percent rejection rate.

Table 7: Null rejection rates at nominal 5 percent size.

dgp	date_iid	hac_delta	moving_block	romano_wolf	row_naive	stationary
null_dep	0.094	0.066	0.070	0.068	0.352	0.067
null_garch	0.098	0.063	0.070	0.067	0.351	0.068
null_iid	0.060	0.060	0.061	0.060	0.062	0.062

decisions from the date-return series. When row-naive rejection is high in null or near-null designs but the dependence-aware methods remain near nominal size, the apparent extra power is a false precision gain from treating same-date rows as independent evidence. The technical appendix reports the full DGP grid, coverage table, power table, and power-curve figure.

The null-size figure and table are the core simulation results: under dependent nulls, row-naive testing rejects about one third of the time at a nominal 5 percent level, while date-level HAC, block-bootstrap, stationary, and Romano-Wolf procedures stay much closer to size.

7 Discussion and limitations

The results should be read as a financial-econometrics testing exercise for entity-time panel evidence. The framework addresses two common sources of false precision: same-date cross-sectional row multiplication and serial dependence in the date-level return series. It does not remove the need for economic interpretation, out-of-sample validation, liquidity analysis, or venue-specific execution modeling.

The formal validity statement assumes a fixed and finite researcher menu, stationary date-return series, adequate moments for the Sharpe delta-method expansion, and block-bootstrap regularity. The framework does not cover long-memory processes, infinite-variance returns, growing researcher menus, structural breaks, or nonstationary date returns without additional assumptions. The empirical stationarity, holdout, factor-alpha, placebo, and cost outputs are therefore diagnostics and boundary checks, not proofs that every maintained assumption holds in every market setting.

8 Data and code availability

The reproducibility package contains the LaTeX source, generated public aggregate artifacts, public-data loaders, simulation code, unit tests, release scripts, and SHA256 provenance. The public repository is: `https://github.com/DsChauhan08/dependence-aware-evaluation-of-panel-trading-strategy-evidence`

Code is released under Apache-2.0 and manuscript/public aggregate artifacts under CC-BY 4.0 unless a target venue requires different terms. Raw third-party data are not redistributed where source terms do not permit redistribution; the repository instead provides documented loaders, source links, access metadata, generated aggregate artifacts, and checksums needed to reproduce the analysis. Proprietary predictions, raw trades, private tickers, production feature definitions, and unrelated workspace artifacts are excluded. The technical appendix contains the full simulation grid, robustness tables, positive-control artifact, registry notes, and proof details omitted from the main manuscript for length.

9 Conclusion

This paper provides a dependence-aware testing protocol for Sharpe-ratio inference in entity-time financial panels. It moves claims from selected symbol-date rows to date-level portfolio returns, estimates Sharpe uncertainty with HAC and dependent-bootstrap methods, corrects the fixed researcher menu, checks same-date placebo and factor-alpha boundaries, and reports turnover-scaled cost sensitivity. The Sharpe UVIF converts the familiar design-effect problem into a variance ratio for the nonlinear Sharpe functional, and ERIR reports how much nominal row-level information remains nonredundant under the equicorrelation diagnostic approximation. For researchers and allocators, the practical implication is direct: statistical significance earned by multiplying dependent rows is not by itself portfolio-performance evidence.

The public campaign shows how this claim boundary changes interpretation. The canonical French WML candidate validates the pipeline against the direct winner-minus-loser construction. By contrast, broader rank-threshold and dynamic Size/BM panel claims fail dependence-aware, placebo, researcher-menu, or cost boundaries; in the Size/BM stress panel, an ERIR of 0.111 implies that about 89 percent of nominal row-level information is redundant under the diagnostic approximation. Pre-aggregated AQR factor series remain useful time-series benchmarks, but without row-level constituent signals they cannot clear the panel permutation gate.

The protocol is therefore a reproducible testing boundary rather than a strategy generator. It helps separate benchmark validation from exploratory search and makes the cost of dependence explicit before a result is promoted as performance evidence. Future work should extend the framework to nonlinear market-impact and capacity models, time-varying cross-sectional dependence, and richer blocked-permutation designs.

Author contributions

Dhananjay S. Chauhan is the sole author of this manuscript and was responsible for conceptualization, methodology, software, validation, formal analysis, data curation, writing, and revision.

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Conflict of interest

The author declares no conflict of interest.

Ethics statement

This study uses publicly available aggregate financial data and simulation data. It does not involve human participants, personal data, clinical data, animal subjects, or cell lines. Ethics approval was therefore not required.

Data availability

The data used in this study are derived from publicly available financial datasets, including Kenneth French data and AQR factor data. Cleaned public aggregate artifacts, source metadata, checksums, and reproduction instructions are available at: `https://github.com/DsChauhan08/dependence-aware-evaluation-of-panel-trading-strategy-evidence`

Raw third-party data are not redistributed where source terms do not permit redistribution.

Code availability

The code used to generate the empirical results, Monte Carlo simulations, tables, figures, and manuscript artifacts is available at: `https://github.com/DsChauhan08/dependence-aware-evaluation-of-panel-trading-strategy-evidence`

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